National University
Of Computer and Emerging Sciences

Foundation of Advancement
Of Science and Technology

Lab Task:	06
Course Code:	CL2005
Course Title:	Database Systems - Lab
Semester:	Spring 2026
Instructor:	Muhammad Mehdi
Email:	muhammad.mehdi@nu.edu.pk

Guidelines

- Please review the lab manual before starting the task to ensure you fully understand the requirements.
- You may use **only the concepts covered in the lab manual** to complete this task.
- Your submission must be a **single PDF file** that includes:
  - The source code (in plain text)
  - Screenshots of the corresponding outputs
- Name your file using the following format: **rollno_name_labtasknumber.pdf**
  Example: *24p-1234_ozair_labtask06.pdf*
- Unethical use of AI tools will result in a **deduction of 5 marks**.
- Late submissions will incur a penalty of **-1 mark per day**.

Tasks

Perform the given tasks using **MySQL** only.

You are required to design a small database schema for a **Book Publishing and Catalog System** consisting of two related tables.

First **connect to the MySQL server from the XAMPP shell** using the `mysql` command (as described in the lab manual).
After connecting to the server, execute all the following MySQL statements inside the server:

1. Create a database named *`your_roll_number`*.
2. Display all available databases in the MySQL server.
3. Select the *`your_roll_number`* database for use.
4. Create a table named `publishers` with the following attributes by determining the suitable data types and constraints as hinted:
   a. `publisher_id`: unique identifier for each publisher that automatically increments.

   b. `name`: name of the publishing company (should not be NULL).
   c. `founded_year`: year the publisher was established.
   d. `website`: official website URL of the publisher (must be unique).
   e. `annual_revenue`: estimated yearly revenue of the publisher.

5. Create a table named `books` with the following attributes by determining the suitable data types and constraints:
   a. `book_id`: unique identifier for each book that automatically increments.
   b. `title`: title of the book (required).
   c. `isbn`: international book identifier which must be unique.
   d. `price`: price of the book.
   e. `publication_date`: date when the book was published.
   f. `page_count`: total number of pages in the book.
   g. `publisher_id`: references the publisher that published the book using named constraints.

6. Display the structure of both tables to verify that they were created correctly.

7. Add a column `genre` to the `books` table to store the genre/category of the book.

8. Alter the column `page_count` in the `books` table so that it must always store a positive value.

9. Add a constraint to ensure that the `founded_year` in the `publishers` table cannot be earlier than the year 1800.

10. Finally verify the updated table structures after performing all `ALTER` operations by displaying the structure of the tables again.